Latent-Function Video Diffusion: Generate Once, Query Videos at Any Continuous Control

Supplementary Material

A. Implementation and Evaluation Details

Family representation. The active encoder is a dense temporal–control mixer, not a control-wise attention-pooling module. The observed controls are sorted by their known coordinates before encoding. Their generated-frame latents receive a channel-wise control embedding with width 128 and scale 0.25, and are placed along an extended temporal axis. A learned square matrix mixes that entire axis. VAE channels and spatial coordinates are preserved. The generated state is one tensor with a locked reference prefix; no group of output frames is required to equal a particular control video.

The direct query decoder uses a scalar hypernetwork with hidden width 128 and four residual blocks containing layer normalization, SiLU, and linear layers. Its output generates a temporal weight matrix of size $4 \times J$ and a bias of size 48×4, where $J = 36$ for lighting and $J = 12$ for MEAD. The matrix is scaled by $J^{-1/2}$ and applied to the sampled code suffix at each channel and spatial location. The same observed reference is prepended to every queried latent. This decoder is continuous in its scalar input; its output need not be monotonic under an external semantic metric.

Optimization. The codec uses AdamW at learning rate 2×10^{-4}, zero weight decay, and cosine decay to 2×10^{-5}. Training uses one complete family per batch and 8,000 steps, with validation every 500 steps. The selected codec is fixed before the flow model is trained. DiT adaptation uses LoRA rank 64 on attention query, key, value, output, and two feed-forward projections, learning rate 10^{-4}, zero weight decay, and training seed 1234. The number of optimizer updates is matched between LFVD and Pointwise, not the number of processed latent tokens. Sampling uses three fixed seeds, 50 steps, and no extra classifier-free guidance evaluation.

Pointwise control conditioning. For Lighting, the strengthened Pointwise baseline encodes s as $[s, s^2, s^3, s^4, \sin(\pi s), \cos(\pi s), \sin(2\pi s), \cos(2\pi s)]$. Two zero-initialized projections inject this vector as a latent bias and as an additional 4096-dimensional context token through an 8–128–4096 MLP. The design is selected on L0 validation before the five test folds are evaluated. It adds 529,968 parameters to the 161,218,560 LoRA parameters; the LFVD codec has 10,493,040 parameters. On L0 validation the token changes Pointwise from .4514 to .3917 LPIPS and raises ρ from .612 to .951. Polynomial–Fourier features were also tested in the LFVD query hypernetwork; exact-code metrics improved slightly, but sampled metrics were unchanged (.336514 versus .336509 LPIPS and identical ρ). We therefore retain the scalar LFVD query.

LFVD-Identity protocol. LFVD-Identity fixes the main-paper family mixer to $\lambda = 0$ and $P_\phi = I$. It uses the same measured families, frozen VAE, Video DiT, rank-64 LoRA targets, optimizer, update count, sampling steps, and random seeds as LFVD-Mix. Its code is formed by directly packing the common reference and all generated-frame latents in control order. It removes the learned coordinate embedding and square family mixer, but trains the same scalar-conditioned reader with latent L_1 reconstruction. The full-grid comparison uses all measured controls during codec and flow training. In the sparse-control comparison, both LFVD-Identity and LFVD-Mix see only controls $\{0, 4, 8\}$ and are evaluated at all nine official light positions. Thus the variant tests family-valued generation and learned post-hoc querying without learned family-coordinate mixing; it is not a Pointwise sampler. Every full-grid method is evaluated on the same locked test records in each fold; sparse-control methods are also re-evaluated on the same locked test records. Separate validation records agree closely.

Lighting coordinates. The nine measured controls have official light IDs $82, 81, 66, 64, 62, 61, 43, 60, 77$, respectively. Their normalized arc coordinates are

$$\{0.0000000, 0.1335363, 0.2278718, 0.3818399, 0.5294032, 0.6233193, 0.7665081, 0.9198963, 1.0000000\}.$$

Each object contributes four camera windows. The five cyclic folds test object groups 1–4, 5–8, 9–12, 13–16, and 17–20, with the next group used for validation and the remaining three groups used for training. Each object is tested once.

MEAD folds. The actor-disjoint test groups are (M003, M005), (M007, M009), (M011, W036), and (W039, W040). Validation actors are M007, M011, W039, and M003, respectively; all remaining actors form the corresponding training split. Each actor contributes the angry and happy emotions at utterances 001, 002, 003, 007, 008,

Table S1. Paired LFVD-Mix minus shared-noise Pointwise differences. Lighting uses the strengthened control-token Pointwise. Intervals use 10,000 object/actor-cluster resamples. Negative LPIPS and positive values for the other metrics favor LFVD.

Dataset	Metric	Difference	95% interval
OpenIllumination	LPIPS	-0.0414	[-0.0511, -0.0316]
	PSNR	+1.363	[+1.139, +1.610]
	SSIM	+0.0208	[+0.0166, +0.0253]
	ρ	+0.0023	[-0.0208, +0.0243]
MEAD	LPIPS	-0.0006	[-0.0151, +0.0187]
	PSNR	+0.428	[-0.086, +0.926]
	SSIM	+0.0063	[-0.0094, +0.0194]
	ρ	-0.0238	[-0.1176, +0.0640]

Table S2. **Dense-training exposure audit** on the scene-disjoint L0 validation fold (25 steps, seed 100). Pointwise receives nine times as many updates. This is a stress test, not the main matched-update protocol.

Method	LPIPS↓	PSNR↑	SSIM↑	ρ ↑
Pointwise, 9× updates	**.3268**	20.78	.7413	**.9979**
LFVD, 1× updates	.3365	**20.86**	**.7487**	.9729

009, and 015. The evaluated artifacts use the final audio-DTW-aligned family release. All methods consume the same aligned family targets and the same reference. Quantitative fidelity uses only the three measured levels; a dense slider visualization would not add new measured targets.

B. Aggregate Uncertainty and Response Diagnostics

We recompute paired intervals from the final per-family export, grouping by object for OpenIllumination and actor for MEAD. For each metric and method, we first average the families and three seeds within a split unit. We then resample the 20 objects or eight actors with replacement 10,000 times and compute the paired difference to shared-noise Pointwise. The 2.5th and 97.5th percentiles form the reported interval. This preserves within-unit dependence and does not treat camera windows, utterances, or sampling seeds as independent subjects. The intervals condition on the fitted checkpoints; they do not include variation from retraining the models.

Table S1 shows a consistent Lighting improvement in LPIPS, PSNR, and SSIM against the strengthened shared-noise Pointwise, while the response-correlation interval includes zero. For MEAD, the intervals include zero for all four metrics, so we report competitive mean fidelity without asserting a statistically resolved advantage. Only eight actors contribute to these MEAD intervals.

The nine-times-update Pointwise run is evaluated only on the scene-disjoint L0 validation fold. Its update multiplier exceeds the family size and is a deliberately strong exposure stress test, not a replacement for the locked five-fold matched-update protocol. Pointwise wins LPIPS and ordering; LFVD wins PSNR and SSIM. The result prevents attributing the main-table fidelity gap solely to representation design.

Table S3. **LFVD-Identity minus LFVD-Mix, five Lighting folds.** Intervals use 10,000 paired bootstrap draws over 20 held-out objects. Positive LPIPS and negative PSNR, SSIM, or ρ favor LFVD-Mix.

Protocol	Metric	Mean	95% CI low	95% CI high
Full grid	LPIPS	-.0014	-.0058	.0030
	PSNR	-.0043	-.0943	.0853
	SSIM	-.0019	-.0040	.0002
	ρ	-.0040	-.0223	.0182
Sparse unseen	LPIPS	.0007	-.0015	.0028
	PSNR	-.0505	-.0998	-.0021
	SSIM	-.0009	-.0021	.0002
Sparse all	ρ	.0013	-.0037	.0069

Table S4. **Encoder instances on four actor-disjoint MEAD folds.** Means cover 112 held-out families, three seeds, and 50 diffusion steps. The final row reports LFVD-Identity minus LFVD-Mix with a 95% paired actor-bootstrap interval.

Method	LPIPS↓	PSNR↑	SSIM↑	ρ ↑
Pointwise-SN	.1157	22.54	.7534	.743
Endpoint interp.	.1152	22.91	.7624	**.859**
LFVD-Mix	.1151	22.97	.7597	.719
LFVD-Identity	**.1069**	**23.16**	**.7656**	.683
Δ Identity−Mix	−.0082	+.19	+.0059	−.036
95% CI	[−.0204, −.0002]	[.00, .46]	[−.0003, .0158]	[−.097, .018]

Table S5. **Target-relative query-path consistency on Lighting.** Errors compare each generated nine-query path with its measured target path; lower is better. Results use 80 held-out families and seed 100.

Method	Adj. LPIPS MAE	Path log err.	Jerk log err.
Pointwise independent	.1190	.5100	.4712
Pointwise shared	.0867	.5866	.5176
Endpoint interpolation	.1439	1.7460	2.1358
LFVD	**.0303**	**.1341**	**.1588**

Table S3 separates the general family-valued benefit from encoder effects. Native-length full-grid and sparse test results are unresolved overall; sparse unseen PSNR has a tiny paired advantage for LFVD-Mix. The raw per-family and per-seed values remain in the accompanying result package.

Table S4 provides the corresponding dynamic-video comparison. LFVD-Identity improves mean LPIPS and PSNR over LFVD-Mix, with paired actor-bootstrap intervals excluding zero, while SSIM and control ordering are unresolved. This reinforces the interpretation of direct packing as a strong family-valued encoder.

The target-relative path metrics use the official nine-control Lighting targets and one common sample seed. Adjacent LPIPS error is the mean absolute difference be-

Table S6. **Reader ablation on the same sampled sparse family codes.** The learned reader and adjacent linear interpolation receive identical LFVD- Identity codes. Results cover five folds, three seeds, and 50 steps.

Reader	LPIPS↓	PSNR↑	SSIM↑	ρ ↑
Adjacent linear	.4074	20.11	.7319	**.9918**
Learned R_ψ	**.3269**	**21.20**	**.7499**	.9902

tween generated and measured adjacent-query LPIPS. Path and jerk errors are absolute log ratios of total adjacent LPIPS length and RMS second-order RGB variation, respectively. Consequently, a collapsed path has large error even if it is smooth. LFVD-minus-shared-Pointwise paired intervals are $[-.0621, -.0506]$, $[-.5304, -.3789]$, and $[-.4489, -.2792]$ for the three columns in Table S5.

The reader ablation reuses exactly the same sampled LFVD-Identity family code for both rows. The adjacent baseline selects the two neighboring packed latent segments and linearly interpolates them. The learned reader improves LPIPS by .0805 (95% CI $[.0488, .1132]$), PSNR by 1.10 dB ($[.11, 2.13]$), and SSIM by .0180 ($[.0016, .0353]$), with nearly identical ordering.

The full sparse-control export includes both observed-control metrics and the relative update budgets. It is the same five-fold, three-seed evaluation as the compact main-paper table; no validation families are included.

Figure S1 provides additional held-out Lighting continua. Within each panel, every row is decoded from the same sampled family code; only the requested physical light coordinate changes.

C. Representation and Training-Family Scaling

The completed scaling study uses nested training sets of 1, 2, 4, 8, 16, 32, and 48 lighting families and the same 16-family L0 validation split. The recorded protocol uses 8,000 codec steps and 3,456 flow updates, a 25-step sampler, and seed 100. The 48-family row reuses the selected full-setting evaluation. The 32-family entry is completed in the final ablation export. These are validation analyses, not additional held-out test results.

The nearly constant exact-code result indicates that reconstruction alone is not a reliable proxy for generative quality in this study. It does not establish an information-theoretic capacity limit or rule out dependence on the coordinate and training schedule. The completed coordinate comparison below shows that the retained nonlinear setting is not uniformly best across validation metrics.

For a fixed trained family representation, increasing the number of requested outputs does not change its denoising trajectory. The native family-code sizes are 7.4 times (lighting) and 2.6 times (MEAD) the number of elements in one ordinary five-frame latent, including their reference prefixes. These ratios describe tensor sizes only. The main paper therefore reports both synchronized wall-clock measurements and a hardware-independent dense-matmul count instead of converting call counts into a claimed speedup.

Timing protocol. All latency measurements use one RTX 4090, resident model weights, BF16, and a 50-step sampler. We synchronize CUDA immediately before and after each timed region, discard setup and warm-up, and average three repeats. The timed region includes the continuous query and frozen-VAE decoding but excludes model loading and file serialization. Pointwise is reported sequentially and with the strongest memory-feasible batch of three. LFVD samples one family code and then evaluates and decodes all requested controls. The reported crossover is fitted from the measured multi-query curve; it is not inferred from trajectory counts.

Analytical transformer work. The FLOP audit uses the locked Wan2.2-TI2V-5B configuration and counts dense matrix multiplications in patch projection, 30 transformer blocks, self-attention and cross-attention projections and attention products, feed-forward projections, text projection, and the output head. One multiply–accumulate is two FLOPs. Layer normalization, elementwise activations, the lightweight query decoder, and VAE decoding are excluded; their device-dependent contribution is represented by the wall-clock test. For latent lengths 5, 13, and 37, the corresponding transformer token counts are 320, 832, and 2,368 at the recorded spatial resolution.

D. Completed Lighting Ablations

All ablations in this section use the fixed L0 lighting validation split. The final export freezes the 3,456-update flow training schedule, a 25-step sampler, seed 100, and the same frozen VAE. The following tables are separate from the three-seed, 50-step object-disjoint main evaluation.

The linear coordinate at scale 0.25 attains sampled LPIPS 0.3242 and response $\rho = 0.9938$, above the retained nonlinear scale-0.25 setting (0.3365 and 0.9729). No claim is made that the nonlinear coordinate is uniquely optimal. Exact-code LPIPS is nearly identical across variants.

Rank-64 LoRA yields lower sampled LPIPS (0.3365) than rank-16 (0.3610) at the recorded training budget. The narrower 24-channel family code is not directly compatible with the frozen 48-channel query interface; no quality metric is reported for that candidate.

Exact-code decoding reaches .0642 LPIPS and perfect ordering, close to the frozen-VAE reconstruction floor of .0512. Sampling the same representation raises LPIPS to

Table S7. **Full sparse-control stress-test results.** Only controls $\{0, 4, 8\}$ are used for training. Pointwise receives nine times the main update budget; LFVD is the unchanged sparse model. Means cover 80 test families, three seeds, and 50 steps.

	Observed controls (3)		Unseen measured controls (6)		Control	Updates
Method	LPIPS ↓	PSNR ↑	LPIPS ↓	PSNR ↑	ρ ↑	relative
Pointwise (control token)	**0.2895**	**21.34**	**0.3322**	20.58	0.9691	9×
LFVD-Mix	0.2993	20.99	0.3410	**21.22**	**0.9906**	1×

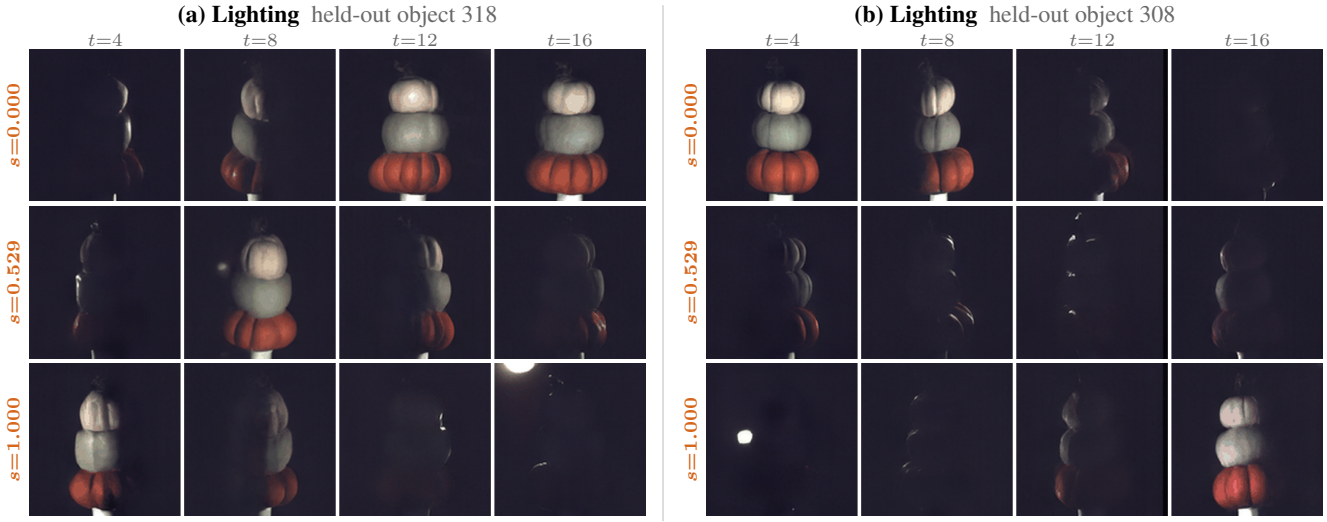


Figure S1. **Continuous control from one LFVD trajectory.** In each panel, all rows are decoded from one sampled family code at different control values s (orange); columns are temporal frames. Both panels show held-out OpenIllumination objects at three measured light coordinates. No query requires an additional Video DiT call.

Table S8. Measured lighting validation scaling rows. Exact and sampled columns report LPIPS; the final column is sampled-code control correlation.

Families	Exact ↓	Sampled ↓	ρ ↑
1	0.0642	0.4500	0.4115
2	0.0642	0.4409	0.4385
4	0.0642	0.4170	0.7948
8	0.0642	0.4021	0.8104
16	0.0642	0.3772	0.9010
32	0.0642	0.3450	0.9542
48	0.0642	0.3365	0.9729

Table S9. Measured generation time (seconds) for M requested videos, averaged over three repeats.

Method	1	3	5	7	9	17	33
Pointwise seq.	**3.88**	11.45	19.06	–	35.38	64.67	127.50
Pointwise batch-3	3.90	**6.62**	**11.16**	16.96	19.86	37.65	72.84
LFVD	13.19	13.93	14.66	**15.40**	**16.15**	**19.14**	**25.11**

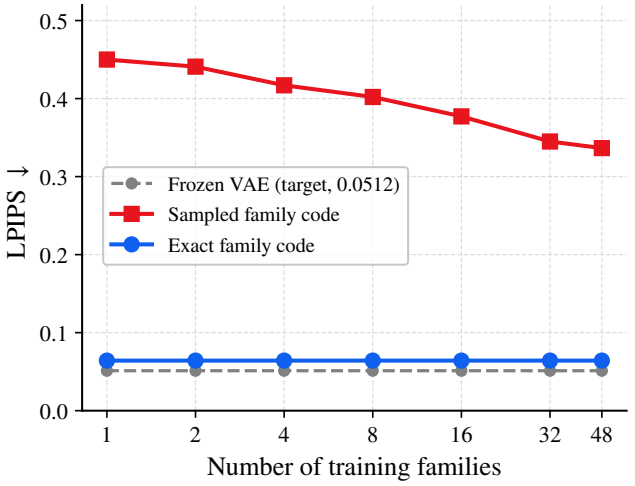


Figure S2. **Completed family-scaling study.** Exact-code reconstruction is nearly unchanged, while sampled-code LPIPS improves as more training families are supplied. This distinguishes family reconstruction from conditional generation under the recorded validation protocol.

.3365, localizing the principal quality loss to diffusion of the family code rather than the query decoder.

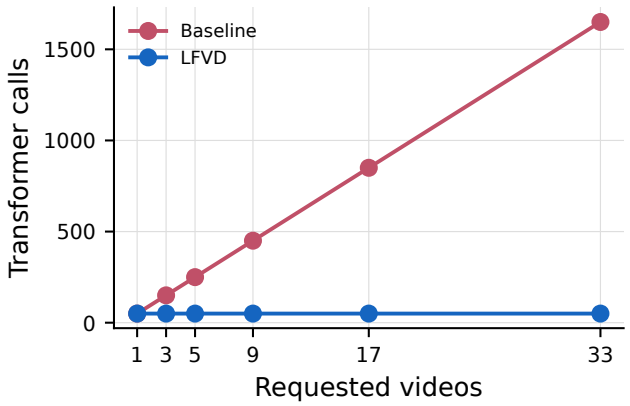


Figure S3. **Call-count scaling, not wall-clock timing.** With a fixed 50-step sampler and no classifier-free guidance branch, Pointwise uses $50M$ transformer calls and LFVD uses 50 family-code calls. The latter process a longer token sequence. Query and VAE costs are not represented on this axis.

Table S10. Analytical transformer cost for 50-step sampling with Wan2.2-TI2V-5B. Query-decoder and VAE-decoder operations are excluded.

Representation	Latent frames	Tokens	TFLOPs/trajectory
Pointwise video	5	320	173.50
MEAD family	13	832	411.52
Lighting family	37	2,368	1,183.59

Table S11. **Coordinate sensitivity on lighting validation.** Exact-code and sampled-code LPIPS use the same decoder. Sampling uses 25 steps and one seed; these values are not the main test results. The main setting is retained without claiming that it wins this sweep.

Coordinate	Scale	Exact ↓	Sampled ↓	ρ ↑
None	–	0.0642	0.3351	0.9531
Nonlinear (main)	0.25	0.0642	0.3365	0.9729
Nonlinear	0.50	0.0642	0.3341	0.9875
Nonlinear	1.00	0.0642	0.3338	0.9771
Linear	0.25	0.0642	0.3242	0.9938

Table S12. **Capacity and LoRA-rank audit** (lighting L0 validation). The 24-channel code is not a runnable configuration of the fixed 48-channel direct-decoder interface; it is an architectural constraint, not a failed quality measurement.

Variant	LPIPS ↓	PSNR ↑	ρ ↑
Code channels = 24	–	–	–
Code channels = 48 (exact)	0.0642	32.06	1.000
LoRA rank 16 (sampled)	0.3610	20.60	0.925
LoRA rank 64 (sampled)	**0.3365**	**20.86**	**0.973**

A non-square LFVD-Compact encoder can change the code's temporal dimension. The 28-frame code retains exact-code LPIPS .0814 and perfect ordering while using 75.7% of the native 37-frame tokens. More aggressive compression preserves order longer than texture: 19 frames still gives $\rho = 1.0$ but degrades LPIPS to .1065. These are representation-oracle results. We therefore run the 28-frame candidate through the unchanged raw flow objective and 50-step, three-seed sampler. Its sampled LPIPS is .4053, PSNR is 21.06 dB, and control correlation is .977. On the same L0 record, full-length LFVD-Mix reaches .3300 LPIPS, quantifying the fidelity cost of the shorter code. Its analytical 50-step DiT cost is 883.87 TFLOPs versus 1183.59 TFLOPs for the 37-frame representation, moving the dense-matmul crossover against a 5-frame Pointwise trajectory from 6.82 to 5.09 queries. These analytical counts do not replace the measured timing experiment in the main paper.

The sparse-control stress test evaluates original measured photographs, never interpolated or model-generated targets. Pointwise uses the frozen control token and 31,104 updates; LFVD uses the unchanged sparse checkpoint. The mixed LPIPS/PSNR result is therefore reported as a trade-off, not an LFVD quality win.

Table S13. **Representation versus generation bottleneck.** Locked lighting validation, 25-step seed-100 setting. The first row is the frozen-VAE reconstruction floor. Exact-code decoding reconstructs the observed family; the last row samples the code from conditions only.

Stage	LPIPS ↓	PSNR ↑	SSIM ↑	ρ ↑
Frozen-VAE target	**0.0512**	**33.17**	**0.8990**	1.000
Exact family code	0.0642	32.06	0.8927	1.000
Sampled family code	0.3365	20.86	0.7487	0.973

Table S14. **Learned-coordinate compression, exact-code oracle.** The 37-frame native coordinate has the same temporal extent as direct packing. Shorter coordinates are trained with the same single reconstruction objective; no perceptual or ordering term is added.

Code frames	Ratio	LPIPS↓	PSNR↑	SSIM↑	ρ ↑
37	1.000	**.0642**	**32.06**	**.8927**	1.000
28	.757	.0814	30.84	.8843	1.000
19	.514	.1065	29.46	.8721	1.000
10	.270	.1750	26.45	.8391	.975

Table S15. **Paired sparse-control stress-test differences** (LFVD-Mix minus nine-times-update control-token Pointwise). Intervals use 10,000 paired cluster-bootstrap draws over 20 held-out objects.

Metric	Difference	95% interval
Unseen LPIPS ↓	+0.0088	[+0.0041, +0.0136]
Unseen PSNR ↑	+0.637	[+0.480, +0.789]
Unseen SSIM ↑	+0.0006	[-0.0020, +0.0030]
Control ρ ↑	+0.0215	[+0.0110, +0.0322]

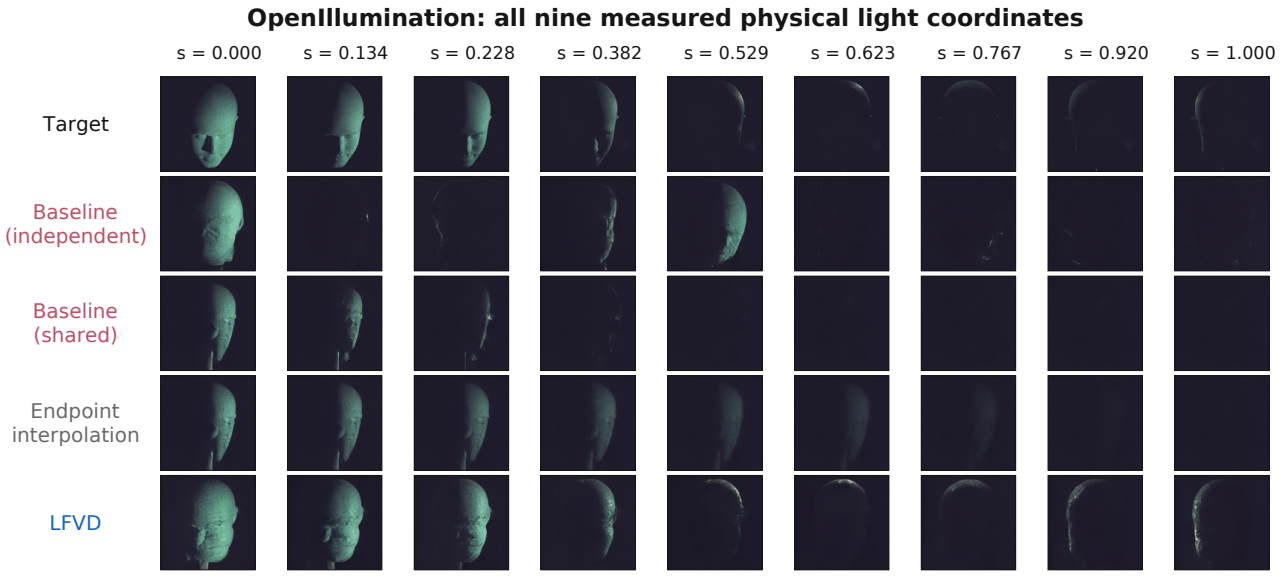


Figure S4. **All measured lighting controls at one camera position.** The fixed lighting example is the source export's median-LPIPS selection in fold L4. Each row uses the same camera index 8. Interpolation is based on two generated endpoints; LFVD's query is learned directly.

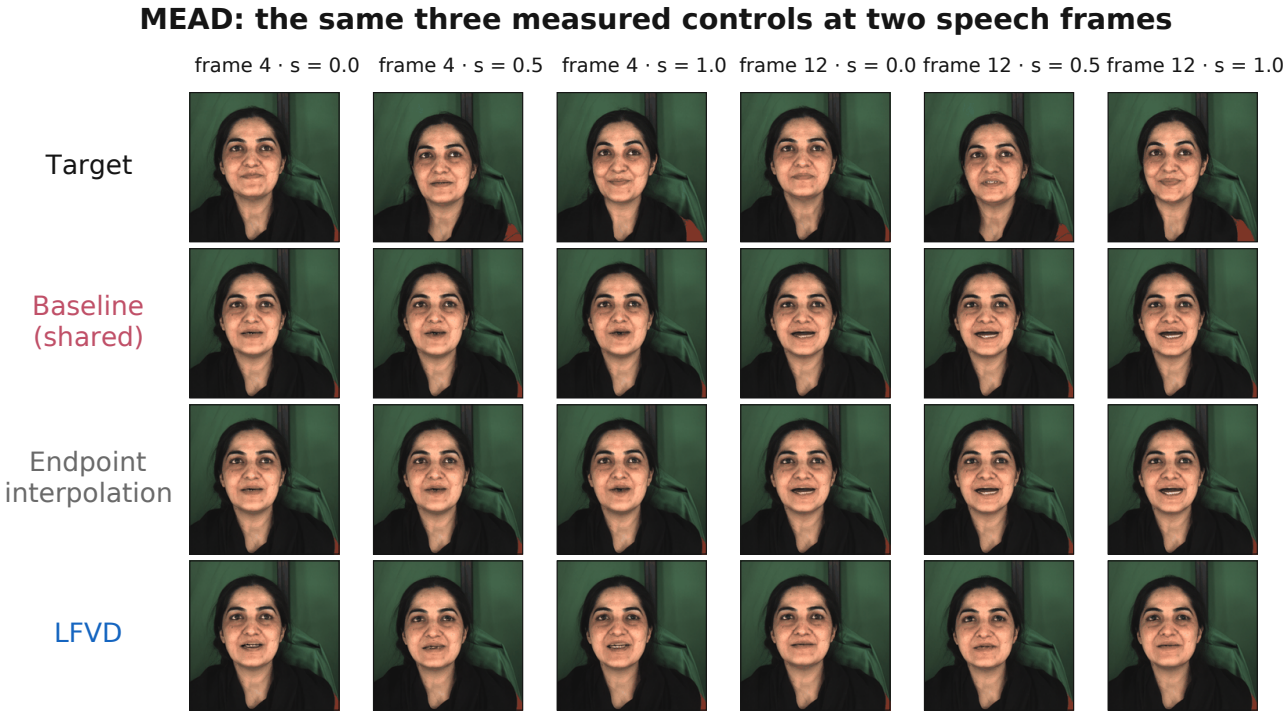


Figure S5. **Native temporal frames at measured expression levels.** Each group of three columns shows the same actor, utterance, seed, and speech-frame position at the three measured controls. The common reference frame is not displayed. These crops show actual outputs and do not serve as ground truth at an unobserved intensity.

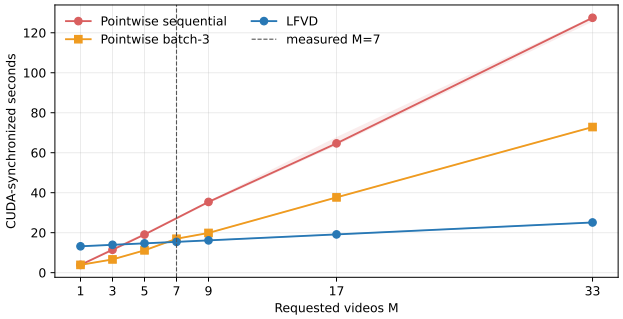


Figure S6. **Measured inference time versus query count.** Timings use 50 denoising steps on one RTX 4090 and average three resident-model repeats. Query and VAE decoding are included. Pointwise batch-3 is the strongest memory-feasible batch; the measured $M = 7$ point confirms the fitted crossover at $M = 6.64$.

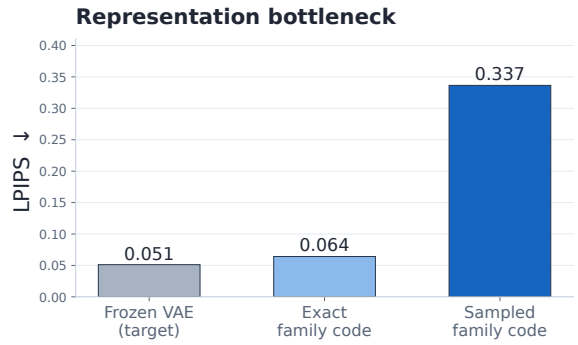


Figure S7. **Bottleneck decomposition** on the locked lighting validation fold: frozen VAE reconstruction, query from an exact family code, and query from a DiT-sampled family code. Lower LPIPS is better; bars show measured averages and are not bootstrap intervals.